Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



Team name: Frustrated Engineers

Team leader name: Nishanth Subramanian E

Theme: BFSI: Intelligent Risk, Fraud & Financial Experiences

Problem Statement: Compliance teams can review only a small sample of RM-customer calls manually. Keyword tools raise false alarms, and plain AI verdicts can't be defended to an auditor. BFSI needs call monitoring at scale where every finding is tied to the exact audio moment, the exact words, and the exact regulation clause.

BRIEF ABOUT THE IDEA
ARGUS | Autonomous BFSI Compliance Intelligence

Turning Every Sales Conversation into Verifiable Compliance Evidence

Financial mis-selling can hide in everyday conversations. ARGUS brings intelligence, transparency, and accountability to financial services by transforming RM–customer sales call recordings into explainable, evidence-backed compliance cases.

Powered by Google Cloud and Gemini AI, ARGUS listens to conversations, investigates potential violations, verifies findings against regulations, and builds an auditable Evidence Chain connecting the exact spoken statement to its timestamp, regulatory clause, and supporting rationale.



From detecting potential violations to proving why they matter — ARGUS empowers compliance teams with traceable evidence for faster, more informed human review.

Opportunities

THE OPPORTUNITY:

- Manual audits cannot efficiently review every sales conversation.
- Keyword-based systems miss context, intent, and nuanced mis-selling.
- AI-generated alerts without supporting evidence create additional work for auditors.

THE VALUE:

- More contextual findings
- Faster evidence verification
- Audit-ready case records

HOW ARGUS IS DIFFERENT:

- **Context-aware detection:** Analyses the meaning of the conversation instead of relying only on keyword matches.
- **Regulation-grounded investigation:** Uses regulatory retrieval to connect findings with relevant SEBI/AMFI provisions.
- **Evidence Chain:** Links the timestamped statement, transcript evidence, regulatory clause, and investigative rationale for auditor verification.
- **Privacy-aware, human-in-the-loop review:** Supports sensitive-data protection and human validation before compliance decisions.

Features Offered by ARGUS

1. AI-Powered Call Transcription
Converts RM–customer call recordings into timestamped, speaker-attributed transcripts.

2. Context-Aware Violation Detection
Identifies potential mis-selling, suitability issues, and disclosure gaps using conversational context.

3. Regulatory Intelligence
Retrieves relevant SEBI/AMFI provisions to ground findings in applicable regulatory requirements.

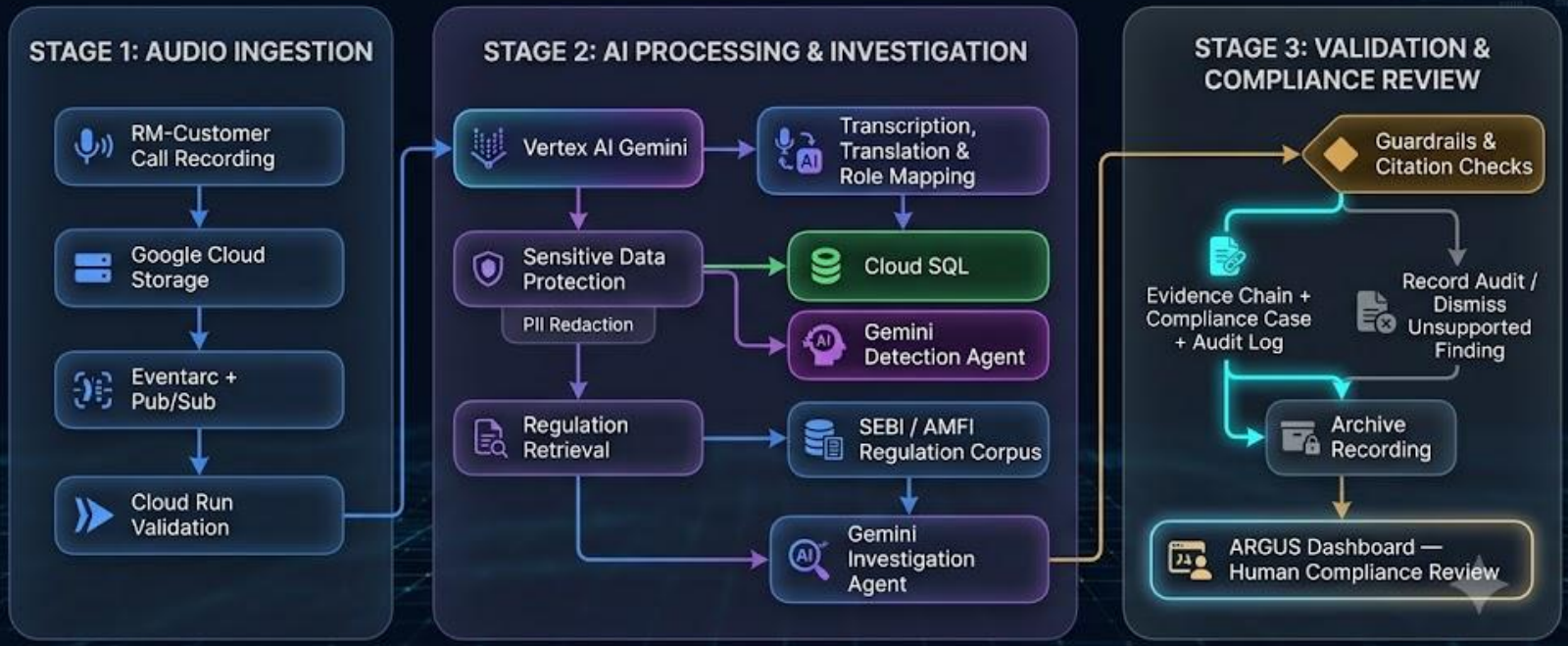
4. Explainable Evidence Chain
Links flagged statements, audio timestamps, regulatory clauses, and investigation rationale for verification.

5. Risk Scoring & Case Management
Organizes findings by risk and provides structured compliance cases for auditor review.

6. Privacy-Aware Audit Workflow
Incorporates sensitive-data protection and human validation to support responsible compliance decisions.

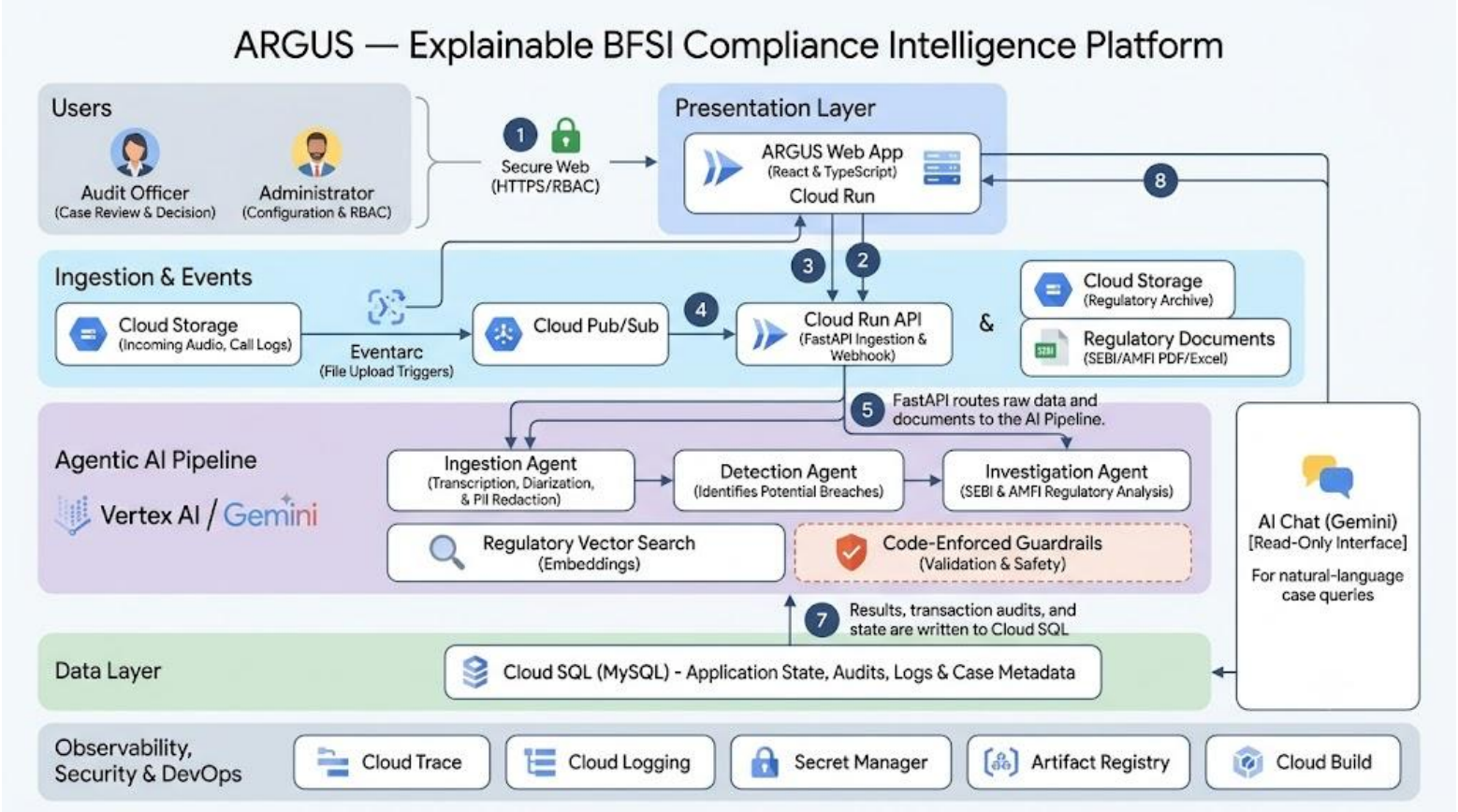
ARGUS — END-TO-END PROCESS FLOW

Enterprise BFSI AI Compliance Intelligence Platform

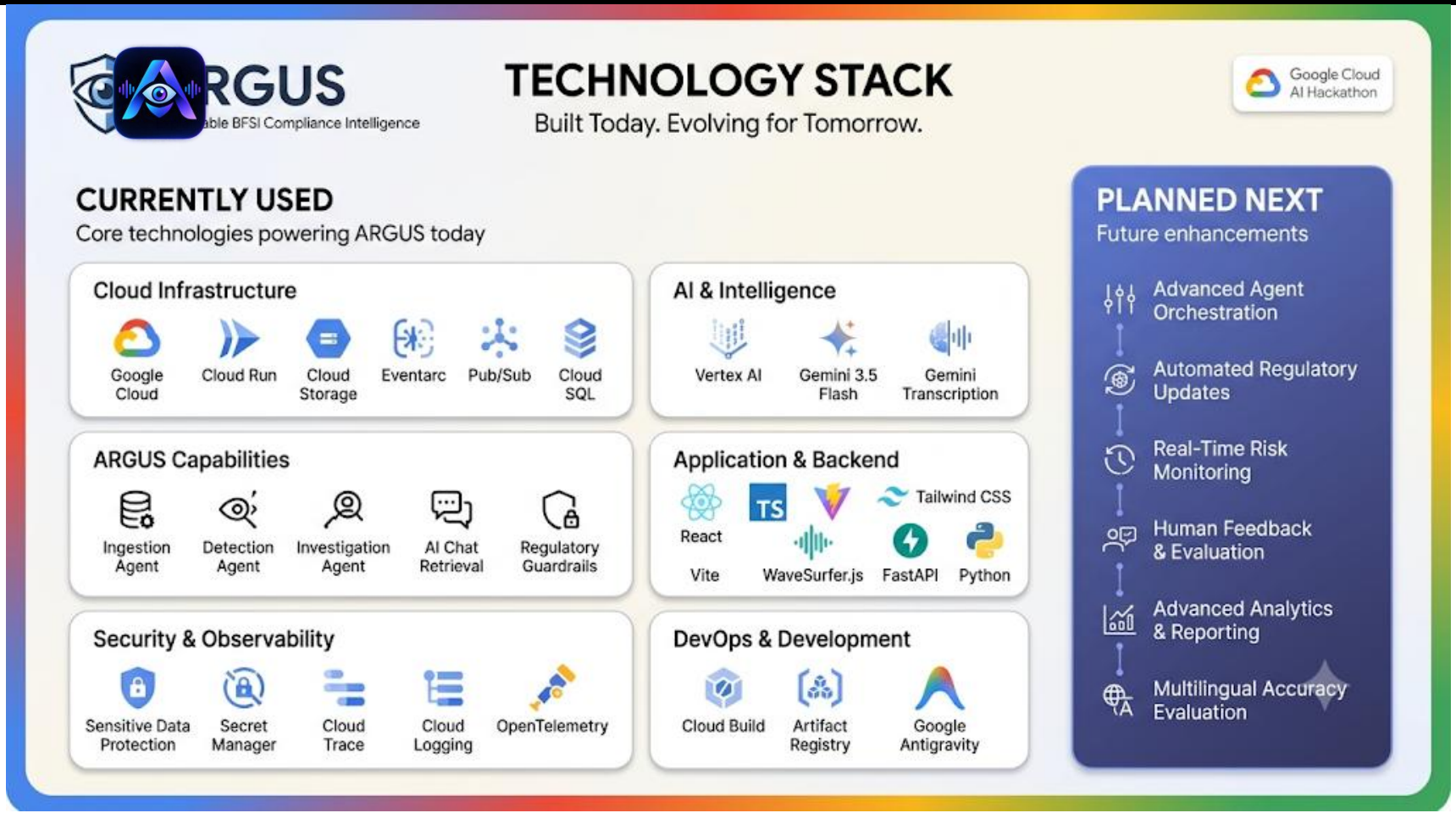


From Call Audio to Explainable, Evidence-Backed Compliance Review

ARCHITECTURE DIAGRAM



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techstack.md

ESTIMATED IMPLEMENTATION COST

Estimated Cloud Operating Cost: ₹4,500–₹13,500 per month

Cost Optimization

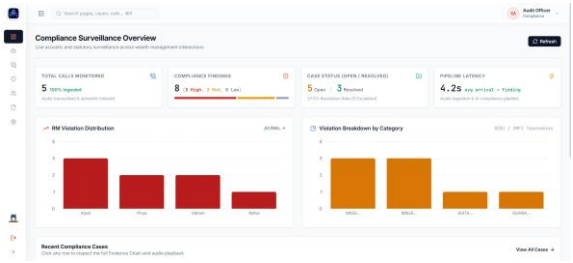
- Serverless, pay-as-you-go cloud infrastructure
- Scalable processing based on workload demand
- Centralized storage and structured evidence management

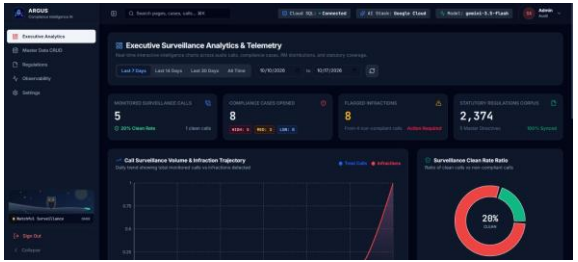
Note: Indicative estimate for a small-scale prototype. Actual costs vary based on usage, model consumption, database configuration, and deployment settings.

Component	Estimated Monthly Cost
Cloud Run — Frontend & Backend	₹500–₹1,500
Cloud SQL — MySQL Database	₹2,000–₹5,000
Vertex AI Gemini — Transcription & AI Analysis	₹1,500–₹5,000
Cloud Storage — Audio & Regulatory Documents	₹100–₹500
Eventarc & Pub/Sub — Event-Driven Processing	₹100–₹500
Logging, Monitoring & Security	₹300–₹1,000

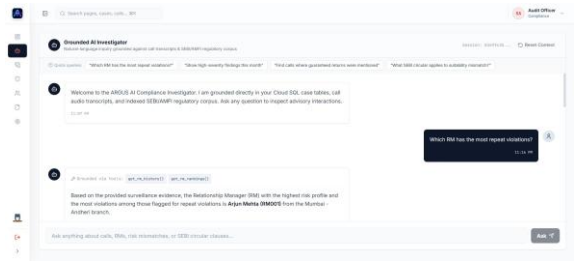
Snapshot Gallery

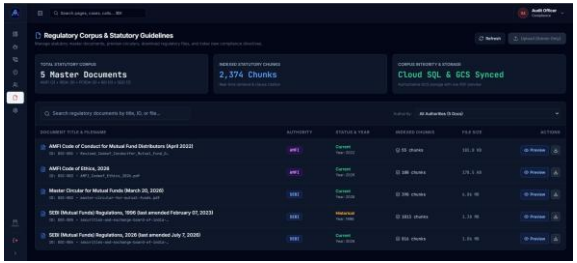


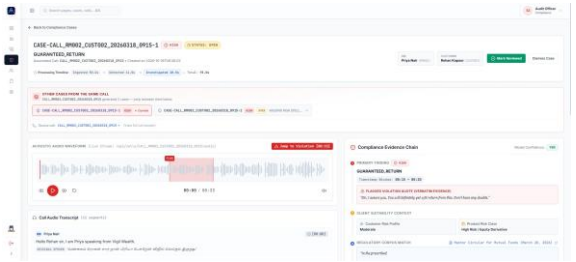




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PROTOTYPE PERFORMANCE & BENCHMARKING

Live end-to-end test on Google Cloud | 10 real call recordings | No mocks

Detection Performance

Approach	Precision	Recall	F1 Score	False Positives
Keyword Baseline	100%	100%	100%	0
Gemini Detection Only	66.7%	100%	80.0%	3
Full ARGUS Pipeline	85.7%	100%	92.3%	1

Key Results

- **100% Recall** — All violations in the evaluated test set were detected.
- **67% Fewer False Positives** — Reduced false alarms from 3 to 1 using Investigation and code-enforced guardrails.
- **88.9% Accuracy** — Reported accuracy for the full pipeline.

Operational Performance

- **158.9 seconds** average end-to-end processing time
- **130.4 seconds** median processing time
- **~$0.23 total estimated cost** for the 10-call benchmark (~$0.02 per call).

KEY TAKEAWAY

Detection alone can over-flag. ARGUS adds evidence-backed investigation by linking findings to regulatory citations, exact transcript quotes, and audio timestamps, helping reviewers validate potential violations.

Note: Results are from a small test set. The keyword baseline performs well on explicit trigger phrases; broader evaluation across varied, paraphrased, and multilingual calls is required. Processing is automated and event-driven, not real-time.

ADDITIONAL DETAILS & FUTURE DEVELOPMENT

ARGUS TODAY (Working Foundation)

- Event-driven Google Cloud architecture with three specialized AI agents.
- Evidence Chain linking audio timestamps, exact transcript quotes, regulatory clauses, and findings.
- Read-only AI Chat for compliance queries, with code-enforced guardrails and observability.
- Validated through live end-to-end testing on real call recordings.

LONG-TERM VISION

Integrate telephony and CRM systems, track regulatory updates, and incorporate human reviewer feedback to evolve ARGUS into an institution-wide compliance intelligence platform

.

TODAY, ARGUS EXPLAINS EVERY FINDING. TOMORROW, IT HELPS PROTECT EVERY CONVERSATION.

FUTURE DEVELOPMENT ROADMAP

01 -> Scale the Intelligence

- Google ADK for coordinated multi-agent orchestration, retries, and escalation.
- Gemini File Search for managed retrieval across expanding regulatory documents.
- Cloud Tasks and Cloud Run Jobs for parallel bulk-call processing.

02 -> Strengthen Trust & Security

- Model Armor for prompt and response screening.
- Cloud KMS and Cloud Storage Bucket Lock for stronger key management and evidence retention.
- Cloud Audit Logs and enterprise identity controls for access governance and traceability.

03 -> Unlock Compliance Insights

- BigQuery and Looker Studio for risk trends across RMs, products, branches, and violation categories.
- Cloud Monitoring and Error Reporting for operational health and alerting.
- Vertex AI evaluation for continuous quality benchmarking and multilingual accuracy assessment.

[ARGUS GitHub Repo Link]

[ARGUS Docs]

[ARGUS App Link]

[ARGUS Demo Link]

Note: For the best viewing experience and clearer visibility of the demo, please watch the video in 1080p (Full HD) or the highest available resolution.

Prototype Demo Credentials

Note: Use these credentials to explore the ARGUS prototype and its role-based features.

Role	Username	Password
Administrator	admin	admin12345
Audit Officer	audit_officer	officer12345

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Thank you

